Review — HyMeKo: A Canonical Hypergraph IR
(IEEE T-SMC submission)

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Note on the journal. The submission directory is named `smc_02`. Here "SMC" denotes the target journal, *IEEE Transactions on Systems, Man & Cybernetics*, and *not* sliding-mode control. The paper is a model-driven-engineering (MDE) contribution; the abbreviation should not be misread.

1 Summary & readiness

HyMeKo is presented as a hub-and-spoke MDE architecture. A single authoritative, signed/typed hypergraph intermediate representation (IR) — built on Levi/Berge incidence with a Blake3 content hash — replaces the usual mesh of pairwise format converters. The IR is manipulated through three operations:

- **compile** (text → IR),
- **project** (IR → sparse tensor),
- **emit** (IR → target text).

The paper claims four algebraic properties, each backed by a proof *sketch* in the appendices:

1. alias invariance,
2. content-addressability,
3. projection–emission independence,
4. bounded storage overhead.

Validation. One IR emits six robot formats (URDF, SDF, Gazebo, MJCF, DOT, Mermaid) at sub-millisecond per robot, reported as 100–700× faster than the `xacro` → `gz sdf` → `mujoco` subprocess stack, alongside a capability gap (the MuJoCo importer fails on long kinematic chains).
State. Late-stage, near submission-ready. The document compiles clean and all 37 bibliography entries resolve. The prior reviewer's hard contradiction on storage overhead ($\rho < 0.7$ asserted in prose vs. $\rho = 2.00$ in the table) has been *fixed*: property A4 now reports ρ improving 2.00 → 1.01.

2 Prioritized findings

3 Technical errors (precise)

1. **Wrong proposition cited.** Cross-view value agreement is attributed to Proposition 3 (projection–emission independence, a cost-factoring statement). It should follow from Proposition 1 (alias invariance) applied per shared query. (P4)

2. **"Idempotent" mislabels a round-trip identity.** $\text{compile} \circ \text{lift} = \text{id}_H$ is a left-inverse / round-trip property, not idempotence ($f \circ f = f$). (P8)

3. $103\times$ **vs** $120\times$**.** The headline speed-up at $|V|$=1000 is arithmetically inconsistent with the paper's own table (34.585 ms $\to$ 4136.6 ms = $120\times$). (P1)

4. **Figure integer inconsistency.** In `fig_pairwise_vs_hub.tex`, caption, drawn box count, and the left-panel grid disagree (9 vs 8 vs 10). (P5)

5. **Unsupported capability-gap claim.** The MuJoCo-FAIL gap (and the $|V|$=5000 "$700\times$") has no supporting in-body table row. (P2)

On the propositions overall. The four "propositions" are honest proof-sketches-by-inspection, and the paper now says so. The storage-overhead proof (A4) is the most rigorous of the four and is correct.

4 Top 3 highest-leverage improvements

1. **Restore Prop. 1's proof to the body** (or inline its lemmas) so the correctness story is self-contained. This is the single most important fix: A2/A3 depend on Prop. 1, so exiling it to supplementary leaves the visible appendices unable to stand alone.

2. **Make every headline number match its table.** Correct $103\times \to 120\times$, and add the $|V|$=5000 and FAIL rows. Quantitative inconsistency erodes trust in *all* the reported numbers.

3. **Align "by construction / impossible" language with what the propositions actually prove**, and fix the Prop. 3 $\to$ Prop. 1 miscitation for cross-view value agreement.

Overall

Near submission-ready. The required changes are surgical fixes, not restructuring. None requires re-running the benchmark — except, optionally, pulling the already-collected $|V|$=2000/5000 CSV rows into the table.

ID	Location	Issue	Severity
P1	`07_eval_scaling.tex` vs `figures/scaling/head_to_head.tex`	Prose "103×" speed-up at $\lvert V\rvert$=1000 contradicts the table (34.585 ms → 4136.6 ms = 120×); inconsistent rounding.	**High**
P2	`tab:head_to_head`	The "700×" ($\lvert V\rvert$=10,5000) and the MuJoCo-FAIL capability gap ($\lvert V\rvert \geq$2000) are claimed in prose, but the table lists only $\lvert V\rvert \in \{10, 100, 1000\}$ — no FAIL row, no 5000 row.	**High**
P3	appendix (Prop. 1 alias-invariance proof)	Prop. 1's proof is exiled to supplementary, yet the A2/A3 proofs explicitly depend on it ⇒ the visible appendices are not self-contained.	**High** (structural soundness)
P4	`05_codegen.tex`, §V, abstract	"Cross-view discrepancy is impossible" cites Proposition 3 (projection–emission *independence*, a cost-factoring claim) — the wrong proposition; value-agreement follows from Prop. 1 (alias invariance) applied per shared query. "By construction/impossible" is overstated.	**High**
P5	`fig_pairwise_vs_hub.tex`	Integer inconsistency: caption "$N+M$=9", panel draws 8 boxes, left 5×5 implies $N+M$=10.	**Medium**
P6	`fig_sota_comparison.tex` (Table V)	The "Approx. impl. surface [LoC/fmt]" column (∼50 HyMeKo vs ∼2000) is marketing-scented and self-admittedly "not a controlled measurement"; soften or drop (also clears a 23pt overfull box).	**Medium**
P7	abstract / intro	Sweep for residual "decidable" phrasing (body already hedged to "practical structural equality"); final-consistency only.	**Low**
P8	`04_hre.tex`	"Compilation is idempotent (compile∘lift = id_H)" mislabels a round-trip/left-inverse identity (idempotence is $f \circ f = f$).	**Low**
P9	presentation	Overfull `\hbox` 23pt in Table V + cosmetic underfull boxes in §VI.	**Low**

Table 1: Prioritized review findings, ordered by severity.